WORKED SOLUTION

STEP I 2007, Question 6

Pure • Diophantine Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q6

The key idea

Both parts begin with a difference of powers, and $x - y$ is a factor of every one of them. Cancelling that factor (allowed because $d \neq 0$) leaves a symmetric expression in x and y. That, together with $x - y = d$, pins down x and y. To get integer solutions we then choose d so that the formulas come out as whole numbers. In part (ii) that means making a square root rational.

Solution

Part (i)

Factorising the left-hand side,

$$(x-y)(x+y) = (x-y)^3.$$

Since $x - y = d \neq 0$, we may divide by $x - y$ to get $x + y = (x-y)^2 = d^2$. Adding and subtracting this and $x - y = d$ gives

$$x = \frac{d^2+d}{2} = \frac{d(d+1)}{2}, \qquad y = \frac{d^2-d}{2} = \frac{d(d-1)}{2}.$$

For the integer pair, put $x = \sqrt{m}$ and $y = \sqrt{n}$. Then $x^2 - y^2 = m - n$, and the required condition is exactly $x^2 - y^2 = (x-y)^3$. With $d = \sqrt{m} - \sqrt{n}$, which is nonzero because $m > n$, we get

$$m = \left(\frac{d(d+1)}{2}\right)^2, \qquad n = \left(\frac{d(d-1)}{2}\right)^2.$$

If d is a positive integer then $d(d \pm 1)$ is a product of consecutive integers, so it is even. Then $\sqrt{m}$ and $\sqrt{n}$ are integers, and so are m and n. Also $m > n$ because $d > 0$. We need $n > 100$, that is $\frac{1}{2}d(d-1) > 10$. The value $d = 5$ gives $\frac{1}{2}d(d-1) = 10$, which is not enough, so we take $d = 6$:

$$\sqrt{m} = 21, \quad \sqrt{n} = 15, \qquad m = 441, \quad n = 225.$$

Check: $m - n = 216 = 6^3 = (21 - 15)^3$. (Other answers, such as $d = 7$ with $m = 784$, $n = 441$, are equally valid.)

Part (ii)

Factorise again: $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Dividing by $x - y = d \neq 0$ gives

$$x^2 + xy + y^2 = d^3.$$

We also have $(x - y)^2 = d^2$, that is, $x^2 - 2xy + y^2 = d^2$. Subtracting this from the previous equation gives $3xy = d^3 - d^2$, as required.

Now substitute $y = x - d$:

$$3x(x - d) = d^3 - d^2 \quad \Longrightarrow \quad 3x^2 - 3dx - (d^3 - d^2) = 0.$$

By the quadratic formula,

$$x = \frac{3d \pm \sqrt{9d^2 + 12(d^3 - d^2)}}{6} = \frac{3d \pm \sqrt{3d^2(4d - 1)}}{6}.$$

Now $\sqrt{3d^2(4d-1)} = |d|\sqrt{3(4d-1)}$, and the $\pm$ already covers both signs, so we may replace $|d|$ by d. Multiplying by 2,

$$2x = d \pm \frac{d\sqrt{3(4d-1)}}{3} = d \pm d\sqrt{\frac{4d-1}{3}},$$

which is the required result.

Ask yourself How do we choose d so that x and y are integers?

If $m = x$ and $n = y$ are integers, then $d = m - n$ is a nonzero integer. For x to be rational, $\sqrt{(4d-1)/3}$ must be rational. Write it as k. Then $k^2 = (4d-1)/3$ is an integer, and the square root of an integer is rational only if the integer is a perfect square, so k is a whole number. Hence we need

$$4d - 1 = 3k^2 \quad \text{for some integer } k \geq 0.$$

The left-hand side is odd, so k is odd. Then $2x = d(1 \pm k)$ is even, so x is an integer, and so is $y = x - d$.

- $k = 1$ gives $d = 1$, so $x = 1$, $y = 0$ or $x = 0$, $y = -1$. Neither pair consists of two positive integers.
- $k = 3$ gives $d = 7$, so $2x = 7 \pm 21$. The plus sign gives $x = 14$, $y = 7$.

So $m = 14$, $n = 7$ works:

$$m^3 - n^3 = 2744 - 343 = 2401 = 7^4 = (m - n)^4.$$

The order matters: with $m = 7$, $n = 14$ the left-hand side is -2401 but the right-hand side is $+2401$.

Checks

- Part (i): $21^2 - 15^2 = 441 - 225 = 216$ and $(21-15)^3 = 216$.
- Part (ii): with $d = 7$, $3xy = 3 \cdot 14 \cdot 7 = 294$ and $d^3 - d^2 = 343 - 49 = 294$.
- A computer search over $1 \le m, n < 200$ finds exactly $(14, 7)$, $(57, 38)$ and $(148, 111)$. These come from $k = 3, 5, 7$ (so $d = 7, 19, 37$), matching the formula. The search also confirms that $(441, 225)$ is the smallest solution in part (i) with $n > 100$.

Common slips

- Forgetting that the cancellation needs $x - y \neq 0$. This is exactly why the question assumes $d \neq 0$.
- In part (i), taking $d = 5$ gives $n = 100$, which fails the strict inequality $n > 100$.
- Writing $\sqrt{d^2} = d$ without comment. It equals $|d|$, and the $\pm$ is what lets us drop the modulus.

Beyond STEP

Part (ii) solves the Diophantine equation $m^3 - n^3 = (m-n)^4$ completely: with $d = (3k^2+1)/4$ for odd $k \ge 3$, the solutions are $m = \frac{1}{2}d(k+1)$ and $n = \frac{1}{2}d(k-1)$. Requiring a square root to be rational is the standard way to turn such an equation into the much simpler condition $4d - 1 = 3k^2$. Equations like this, and the arithmetic of quadratic forms behind them, are treated in Davenport, *The Higher Arithmetic*.

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